

1) A local hospital compiles data on the length of time a patient is in surgery and the length of their stay in the hospital after surgery. A newbie hospital worker notices that the length of time a patient is in surgery is highly correlated with their hospital stay after. Explain to the newbie why this correlation does not imply a causation.

2) The good folks at Apples Inc want to create bags of presliced apples for kids to easily put in their lunch boxes. They want to test different mixtures of preservatives (A, B, and C) on their Honeycrisp apples. They will treat all the apples in a bushel of Honeycrisps by randomly assigning them to each preservation treatment, and then comparing how long they are able to remain in the bag before they begin to brown.

(a) Identify the experimental units, the explanatory and response variables, and the treatments.

(b) The researchers plan to use a completely randomized design. Describe how they should assign treatments to the 126 experimental units.

(c) The researchers suspect that the type of apple will influence how well the preservation method works. They want to repeat this experiment with 4 different types of apples. Describe how they should change the design of the experiment to account for this addition.

3) A study was conducted to determine if taking a daily dose of aspirin reduces chance of catching the common cold. This study was conducted using 550 volunteers who were not already on an aspirin regimen. The subjects were randomly assigned to one of two groups: a treatment group who received a low dose of aspirin daily, or a control group who received a placebo. The subjects were unaware of what group they were in. At the end of the study, the subjects were asked to meet with a doctor to discuss if they had any symptoms pertaining to the common cold.

(a) Is this study an experiment or an observational study? Explain your answer.

(b) What would be the advantage of having this study be double blind?

(c) Would blocking according to gender be worthwhile in this study? Explain your answer.

(d) Describe what the “placebo effect” would look like in this experiment.

4) Type 1 diabetes is thought to be caused by an autoimmune reaction (the body attacks itself by mistake) that destroys the cells in the pancreas that make insulin. It is a disease that currently has no cure. Researchers was to conduct a clinical trial on a promising new drug that help the body produce its own insulin.

(a) You have 400 volunteers with Type 1 diabetes. Describe a completely randomized design for this experiment.

(b) You have 250 children (17 and younger) with Type 1 diabetes (120 females and 130 males), as well as 300 adults (18 and older) with Type 1 diabetes (190 females and 110 males). Describe a completely randomized block design.

(c) Explain why a matched pairs design would not work for this experiment.